



Preparation of fluorinated/silanized polyacrylates amphiphilic polymers and their anticorrosion and antifouling performance

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ABSTRACT

A series of fluorinated/silanized polyacrylates amphiphilic polymers were successfully synthesized by changing its fluorinated groups content (C_{FP}) and then thoroughly characterized. A modified corrosion resistance (R_{sd}) that takes the polymer film thickness into account was proposed to evaluate the anticorrosion performance of the polymers. The results show that the thermal stability of all polymers is high with the decomposition temperature being higher than 330 °C, and the R_{sd} larger than $10^8 \Omega \text{cm}^2 \mu\text{m}^{-1}$ even after immersion in 3.5 wt% NaCl static solution for 7 d. Meanwhile, almost all the polymer's water contact angle, surface smoothness, R_{sd} and anti-fouling performance (except for the adsorption of HFG on the polymer of C_{FP} being ca. 26.7 wt%) increase with the increment of C_{FP} , whereas C_{FP} has negligible influence on polymer's glass transition temperature. Furthermore, the polymer of C_{FP} being ca. 15.4 wt% exhibits the lower adsorption and the maximum removal of both BAS and HFG proteins.

1. Introduction

The accumulation of microorganisms on the surface of materials in an aqueous environment can cause many serious problems, such as biofouling and accelerating corrosion of metallic materials [1,2]. Protein adsorption plays an important role in the process of biofouling [3–5], and a wide attention has been drawn to the polymers with anti-protein adsorption property [6–8]. In recent years, many hydrophilic, hydrophobic and amphiphilic materials have been reported for anti-protein adsorption. It is generally believed that the hydrophilic materials [9–11] based on oligo(ethylene glycol) (OEG), poly(ethylene glycol) (PEG), etc. can form a layer of hydrated film, which can effectively inhibit the adhesion of microorganisms on the surface of materials [12,13]. Whilst the siliconpolymers [14,15] or fluoropolymers [16–18] based hydrophobic materials also possess good antifouling performance due to their low surface energy [19]. Especially, the amphiphilic materials containing both hydrophobic and hydrophilic components possess better protein adsorption resistance than both the hydrophilic or hydrophobic materials and have been attracting great interest [20–23]. Our recent work [24] found that, the surface wettability of the amphiphilic polymers can be regulated by its fluorinated groups content (C_{FP}). What is more, the determinant factor for its

protein adsorption amount gradually turns from the composition of its surface to its internal as the increase of C_{FP} . Furthermore, when the fluorinated degree of the bulk/internal polymer layer (k) is approximately 12, the polymer film possesses the minimal protein adsorption.

For any polymer coatings used in an aqueous environment, one of which important role is to protect the metallic materials against corrosion. Therefore, an ideal antifouling film for industrial applications should also possess excellent corrosion resistance in addition to its good antifouling performance. Unfortunately, although the antifouling property of many amphiphilic polymers has been studied in details, their anticorrosive property on metallic substrates is scarcely explored.

In this study, a series of amphiphilic polymers were synthesized according to our previous work [24] replacing partial monomers for the purposes of simultaneous antifouling and anticorrosion (for metals, such as A3 steel). Composition of the polymers were controlled by varying the content of the fluorinated groups, and the structure, thermal stability, surface wettability, adhesion strength, hardness, anticorrosion and the protein adsorption/release performances of the polymers were thoroughly investigated.

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2. Experimental section

2.1. Materials

Analytical reagents: methyl methacrylate (MMA), n-butyl acrylate (n-BA), styrene (st), xylene (from Sinopharm, Shanghai), diiodomethane (Aladdin) and γ -Methacryloxypropyltrimethoxysilane (γ -MPS, Macklin) were used as received. 2-Hydroxyethyl methacrylate (HEMA, Macklin) and 2-perfluorooctylethyl methacrylate (FMA, Aldrich) were purified according to the literature [5]. Benzoyl peroxide (BPO, Macklin) was recrystallized with chloroform and methanol before used. Bovine serum albumin (BSA, Macklin) and human plasma fibrinogen (HFG, Hefei Bomei Biotechnology Co. Ltd.) were also used as received.

2.2. Preparation of fluorinated/silanized polyacrylates

The preparation procedure of fluorinated/silanized amphiphilic polymers was similar to the literature [24], and a representative preparation procedure was as follows: Xylene (65.0 mL) and γ -MPS (2.0 g, 8.1 mmol) was heated to 75 °C with mechanical stirring. Subsequently, a mixed solution of MMA (24.8 g, 247.6 mmol), BA (15.2 g, 118.6 mmol), st (2.0 g, 19.2 mmol), HEMA (4.0 g, 38.8 mmol) and BPO (0.336 g, 1.39 mmol) was added dropwise within 4 h under a nitrogen atmosphere. Then, FMA (8.0 g, 14.9 mmol) and BPO (0.06 g, 0.25 mmol) were sufficiently dispersed in xylene (15.0 mL), and added to the stirred solution drop by drop in 1 h and then stirred for another 3 h at 75 °C. A series of fluorinated/silanized amphiphilic polymers were obtained using the same method by changing the mass of FMA.

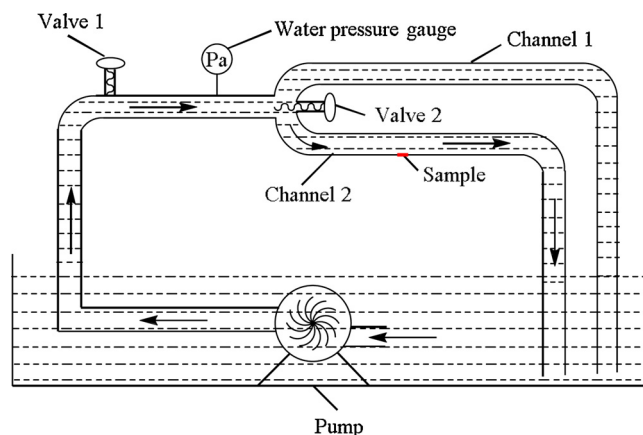
2.3. Preparation of samples for characteristic measurements

The obtained polymers were dried in a vacuum at 120 °C for 12 h for Fourier transform infrared measurement and thermal analyses. 2 g obtained polymers (reaction solvent containing) were distilled under reduced pressure and diluted with 10 g THF. Repeated the "distillation-dilution" process three times to remove the unreacted monomers and reaction solvent. Finally, diluted the polymers in THF to about 0.1 wt% for GPC measurement. Sprayed the polymers on horizontally placed A3 steel substrate according to the literature [25], and the samples were dried at room temperature for at least 3 days for other characteristic measurements.

2.4. Characterization

Fourier transform infrared (FT-IR) spectra were recorded on a Nicolet iS10 spectrometer (KBr disc). Gel permeation chromatography (GPC) was obtained on a D40 (TSK CO., Waters 2414 refractive index detector) at 40 °C (flow rate: 1.0 mL min⁻¹) using tetrahydrofuran (THF) as the eluent and polystyrene (PS) as a standard to calibrate the chromatograms. Thermal analyses were performed on a TA Q50 thermal analyzer. The samples for thermogravimetric analysis (TGA) were heated from room temperature to 800 °C with a heating rate of 10 °C min⁻¹ under a nitrogen atmosphere at a flow rate of 60 mL min⁻¹. Samples for differential scanning calorimetry (DSC) were heated at 10 °C min⁻¹ from -10 °C to 100 °C at a flow rate of 60 mL min⁻¹ (second heating scan).

Contact angles (CAs) were measured by a JC2000 contact angle instrument at room temperature with 5.0 μ L droplets. Surface free energy of the polymers was estimated from the corresponding water and diiodomethane CA according to the literature [26]. Scanning electron microscopy (SEM) was characterized on a SU-1080 at 3 kv. X-ray photoelectron spectroscopy (XPS) was performed using a Thermo Scientific ESCALAB 250Xi with an Al K α X-ray source (1486.6 eV) at a takeoff angle of 30°. Adhesion was obtained by QFH cross-cut tester (Minglang, 1 mm) according to ASTM D 3359-17 Test Method B and



Scheme 1. Schematic diagram of the device for protein release.

hardness was tested by QHQ-A pencil tester (Ruifeng) referring to ASTM D 3363-2005. Thickness was measured by EC-770 (Yuwese). Electrochemical impedance spectroscopy (EIS) was measured on a PARSTAT-2273 at the frequency range of 100 kHz to 10 mHz with an AC amplitude of 20 mV at room temperature. A typical three-electrode system was used in the measurement: A3 steel substrate coated with samples worked as a working electrode, platinum stick and saturated calomel electrode (SCE) worked as counter electrode and reference electrode, respectively. The working electrodes were continuously immersed in an electrolytic tank containing 250 mL 3.5 wt % NaCl static solution, and the solution was not changed until the test was completed.

2.5. Protein adsorption and release

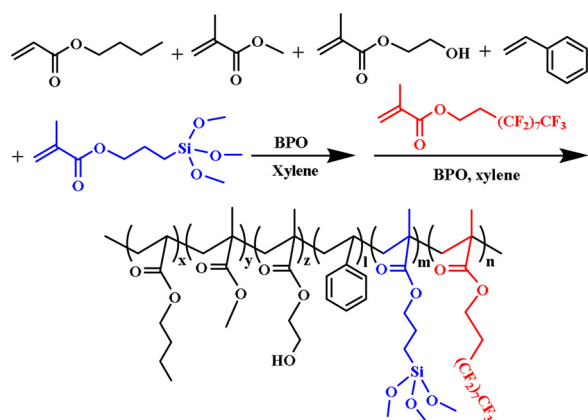
Protein adsorption performance of polymers was evaluated according to the literatures [13,24,27]. Following Schultz's experiment for Amphora desorption [28], we designed a similar device (showed in Scheme 1) to measure the release of proteins. Controlled the flow pressure of channel 1 through Valve 1 to be stable at $\sim$ 85 Pa, and then the Valve 2 was switched to flush the samples in channel 2 for 2 s. A3 steel disc with a diameter of 5 mm was used as the substrate for samples. The samples were then washed with deionized water and dried in a stream of nitrogen. Those samples were subsequently measured by XPS, and At. N_{1s} (%) was calculated by integrating the peak areas of N_{1s} peak.

3. Results and discussion

3.1. Synthesis and characterization of fluorinated/silanized polyacrylates

Polyacrylates were selected as the component of antifouling coatings due to their good corrosion resistance [25,29] and excellent mechanical robustness [30]. Furthermore, fluorinated polyacrylates prepared by doping with fluorinated acrylic monomer can significantly improve the photochemical stability of the polymers [31–33]. Therefore, MMA, n-BA and st were selected as the basic film-forming monomers to ensure the basic properties of the polymers. HEMA was used to enhance the adhesion of polymers, while γ -MPS and FMA acted as functional monomers to reduce the polymer's surface free energy. A two-step free radical initiated polymerization method was employed to prepare the polymers (Scheme 2). By altering the content of FMA, it is possible to change the polymer's composition and structure, and therefore significantly affects its properties. The fluorine-free polymer was designated as SPA, whilst those with FMA content varying from 4.3 wt% to 26.7 wt% were named as FSPAs. The composition and characterization of SPA and FSPAs were list in Table 1.

The structure of the polymers were first confirmed by FT-IR (Fig. 1).



Scheme 2. Preparation of fluorinated/silanized polyacrylates.

Compared to SPA, a new weak peak at 660 cm^{-1} is observed in all of FSPAs, which is attributed to the wagging vibration of C–F, indicating that FMA has been successfully embedded in the polymers. Furthermore, the conversion rate (Conv., %) of monomers (vinyl groups) has been investigated, and the results are listed in Table 1. It shows that all FSPA polymers have high conversion (larger than 90 %), indicating that most of the monomers have been involved in polymerization. GPC measurement shows that FSPAs possess a relatively larger molecular weight and lower polydispersity index (PDI) than SPA (Table 1). The decomposition temperature (T_d) and the glass transition temperature (T_g) of SPA and FSPAs were respectively obtained from TGA and DSC (Fig. 2), and the results were also listed in Table 1.

It is obvious that all the polymers show a one-step thermal decomposition process (Fig. 2a), and possess good thermal stability due to T_d values exceeding 330°C . No crystallization or melting peak is observed for SPA and each FSPAs, except for one glass transition appears in the investigated temperature range from -10°C to 100°C (Fig. 2b), reflecting that these polymers are amorphous. In general, T_g of the polymers should increase significantly with the increase of the FMA content according to Fox Equation [34] since FMA works as a hard monomer. However, the T_g values (Table 1) for SPA and all FSPAs are much close and do not change significantly with the content of FMA. It means that the embedding of FMA does not change the geometry of the polymers. It is well known that methoxysilanes are highly reactive chemicals, and they do cross-link quite easily under essentially. That is to say, a partial network structure is formed between the polymers due to the cross-linking, thus making the molecular chain difficult to rotate.

3.2. Surface characteristics of polymers

The antifouling performance of FRCs is mainly dependent on their surface free energy, elastic modulus and surface roughness [4,35]. Therefore, the surface characteristics of SPA and FSPAs were characterized by contact angle measurements (Fig. 3) and SEM (Fig. 4). Obviously, even the embedded FMA is as low as 4.3 wt%, the WCA of the films is significantly increased from 85.4° (SPA) to 105.4° (FSPA-1),

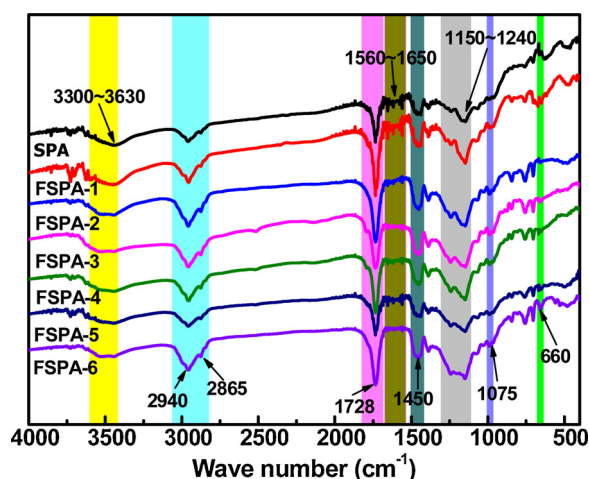


Fig. 1. FT-IR spectra of SPA and FSPAs.

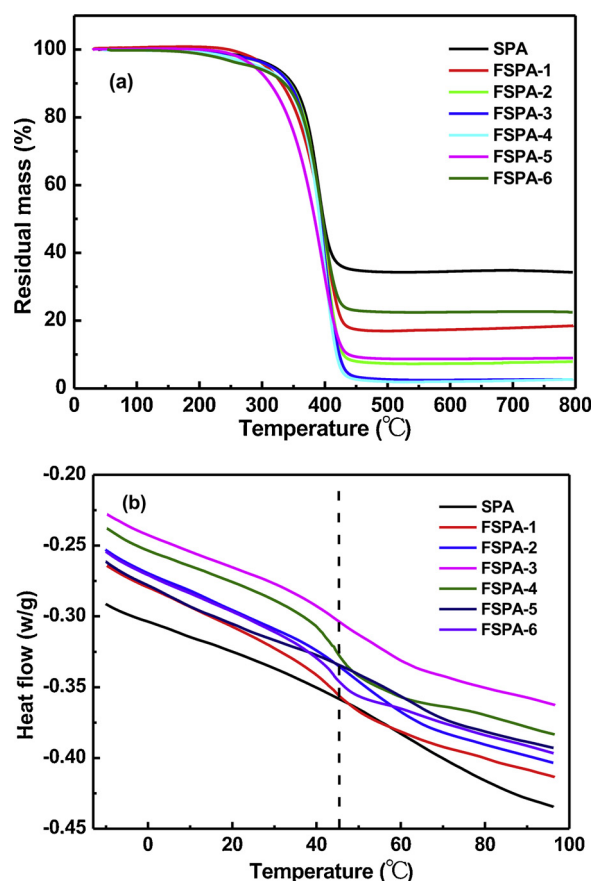


Fig. 2. (a) TGA and (b) DSC curves of SPA and FSPAs.

Table 1

Composition and characterization of SPA and FSPAs.

Sample	MMA/n-BA/st/ HEMA/ γ -MPS (g)	FMA (wt %)	BPO (wt %)	$M_n \times 10^{-4}$ (g/mol)	PDI	T_d ($^\circ\text{C}$)	T_g ($^\circ\text{C}$)	Conv. (%)
SPA	24.8/15.2/2/4/2	0.0	0.7	12.4	3.16	356	44.5	86.0
FSPA-1	24.8/15.2/2/4/2	4.3	0.7	18.6	1.79	344	45.4	98.9
FSPA-2	24.8/15.2/2/4/2	8.3	0.7	17.9	1.83	358	46.4	94.4
FSPA-3	24.8/15.2/2/4/2	12.0	0.7	18.3	1.94	358	44.4	95.3
FSPA-4	24.8/15.2/2/4/2	15.4	0.7	18.8	1.76	360	44.7	92.9
FSPA-5	24.8/15.2/2/4/2	21.4	0.7	19.2	1.84	336	46.9	98.1
FSPA-6	24.8/15.2/2/4/2	26.7	0.7	18.7	2.03	358	44.6	91.8

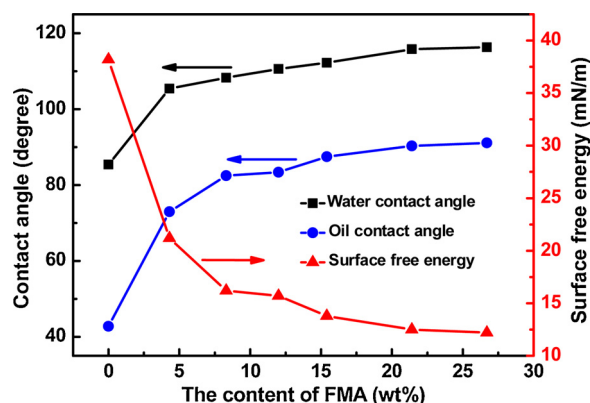


Fig. 3. Contact angles and surface free energies of the polymer films.

while the surface free energy is sharply reduced from 38.2 mN/m (SPA) to 21.2 mN/m (FSPA-1). With the increase of FMA content, WCA of the FSPAs increases to 116.3°, which is much close to the theoretical value of PFMA (118°) [36]. However, detailed inspection indicates that, when the FMA content is larger than 21.4 wt%, the continuous increase of FMA content has no significant effect on both water contact angle and surface free energy of the FSPA films, which has been mainly attributed to the enrichment and saturation of the fluorinated groups on the polymer surface of FSPA films [22,24]. In the field of fluorinated coatings, the key to success is to reduce the amount of fluorinated monomers maintaining excellent performances in terms of water repellency, and many researchers have achieved great success in this area [37–39]. Although the resulting polymers here contain relatively large fluorinated monomers content, FSPA films possess better water repellency, which is unmatched by many of the polymers reported in the literatures [37–39].

The embedding of FMA into SPA can significantly influence their surface morphologies (Fig. 4). It can be clearly seen that some of spherical protrusions are sparsely distributed on the surface of SPA. However, when FMA is embedded, these spherical protrusions are not

Table 2
Adhesion strength and hardness of polymers.

Polymer	Adhesion (grade)		Hardness
	A3 steel	Glass	
SPA	5B	5B	3H
FSPA-1	5B	5B	3H
FSPA-2	5B	5B	3H
FSPA-3	5B	5B	3H
FSPA-4	5B	4B	3H
FSPA-5	4B	4B	HB
FSPA-6	4B	4B	HB

observed. In addition, as the content of FMA increases, the surface morphology of FSPA gradually changes from that with a small number of wrinkles (FSPA-1) to much more smooth (FSPA-4 and FSPA-6), which has been demonstrated to be beneficial for antifouling [35,40–42].

3.3. Adhesion strength and hardness of polymers

Good adhesion strength and hardness of coatings are the precondition for their expected service life. Generally, for industrial applications of coatings, it is satisfactory that the adhesion strength is not less than 3B and the hardness is not lower than HB. Therefore, the adhesion strength (on A3 steel and glass) and hardness of SPA and FSPAs were investigated (Table 2). Due to the high content of HEMA and γ -MPS in the polymer as well as cross-linking, all of the polymers exhibit excellent adhesion to both A3 steel and glass, although the adhesion strength is slightly reduced (from 5B to 4B) when the content of FMA is large (21.4 wt% and 15.4 wt% for A3 steel and glass, respectively). The decrease of adhesion can be mainly attributed to the incompatibility between the good hydrophobicity (reflected by low surface free energy) of FSPAs and the hydrophilic of substrates when the content of FMA is larger. Additionally, when the content of FMA is less than 21.4 wt%, the hardness of the polymers is as high as 3H, which means that these polymers possess excellent hardness for industrial

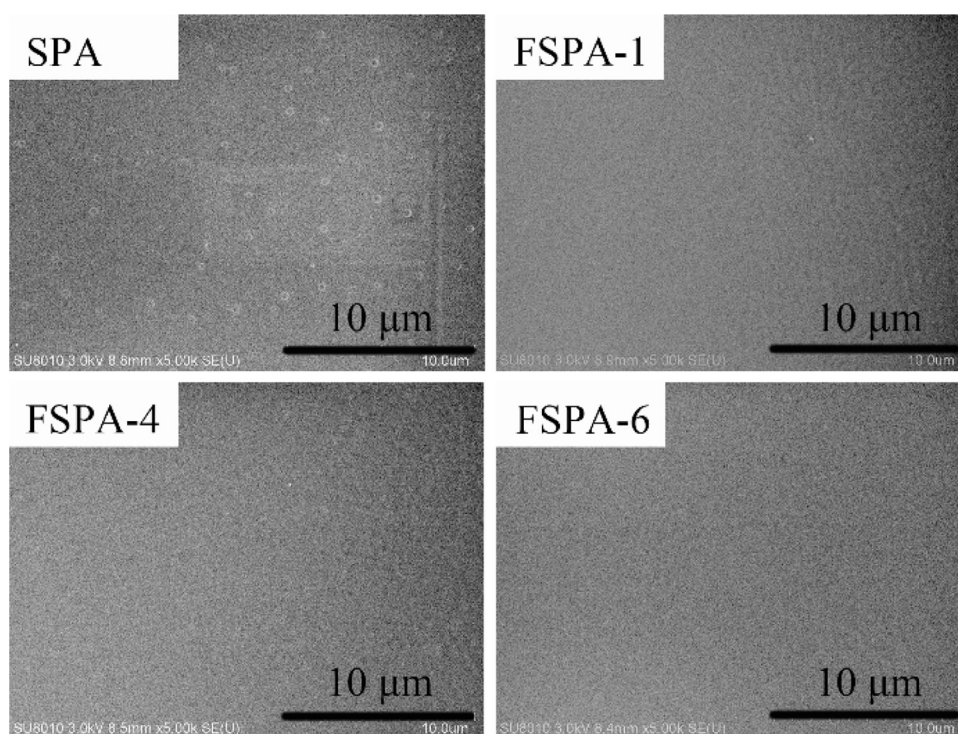


Fig. 4. SEM images of the polymer films.

Table 3
Stability of the samples in salt solution.

Sample	Working area (S , cm^2) ^a	Thickness (d , μm)		Contact angle after immersion ($^\circ$)	Water absorption (%) ^b
		Before immersion	After immersion		
SPA	1.54	80.2	85.3	82.7	1.77 %
FSPA-1	1.54	88.6	91.8	103.6	1.23 %
FSPA-4	1.54	32.4	36.2	111.0	1.06 %
FSPA-6	1.54	24.8	35.7	109.3	0.88 %

^a Working area is the area of working electrode in EIS.

^b Water absorption is calculated based the equation: $\text{Absorption} = \frac{m_2 - m_1}{m_1 - m_0} \times 100\%$, where m_0 is the weight of substrate, m_1 and m_2 are the weight of substrate coated with the polymer before and after being immersed in 3.5 wt% NaCl static solution for 7 days, respectively.

applications.

3.4. Stability and corrosion resistance of polymers in static salt solution

Good stability and anticorrosion performance of coatings are also the guarantee for their industrial application, especially in marine environments. Table 3 lists the thickness, water contact angle and water absorption of the samples before and after being immersed in 3.5 wt% NaCl static solution for 7 days (the contact angle data before immersion is plotted in Fig. 3). It is obvious that all of the polymers possess a very low water absorption and which gradually decreases with the increase of FMA content, and the thickness increment of any polymers is less than $5\mu\text{m}$ after immersion except for FSPA-6. Thanks to the cross-linking between the polymer molecules, the formed film is denser and

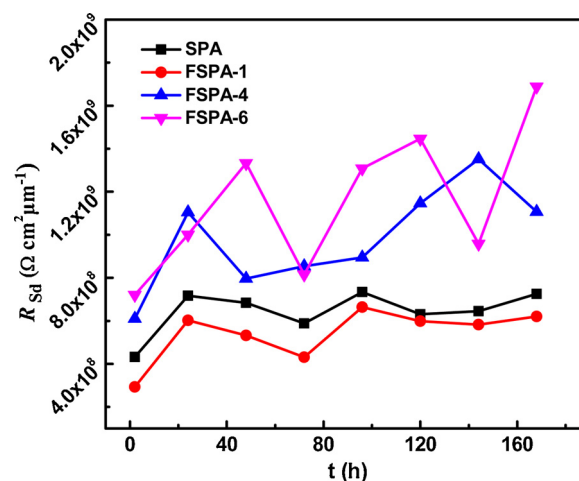


Fig. 6. Evolution of R_{sd} with immersion time.

thus the film has a very low water absorption and thickness increment. In addition, more fluorinated groups are enriched on the surface of the film when FMA content in the polymer is larger, which enhances the hydrophobicity of the film. Therefore, it is difficult for water to penetrate into the film, thus reducing the water absorption. Surface rearrangement has been demonstrated to be the main reason for the loss of hydrophobic properties of fluorinated polymers [26,43,44]. It is worth to mention that after being immersed in 3.5 wt% NaCl static solution for 7 days, FSPAs still have good hydrophobicity, especially for FSPA-4, the contact angle reduces only 1.2° . It indicates that the cross-linking between polymer molecules seems to “freeze” the surface

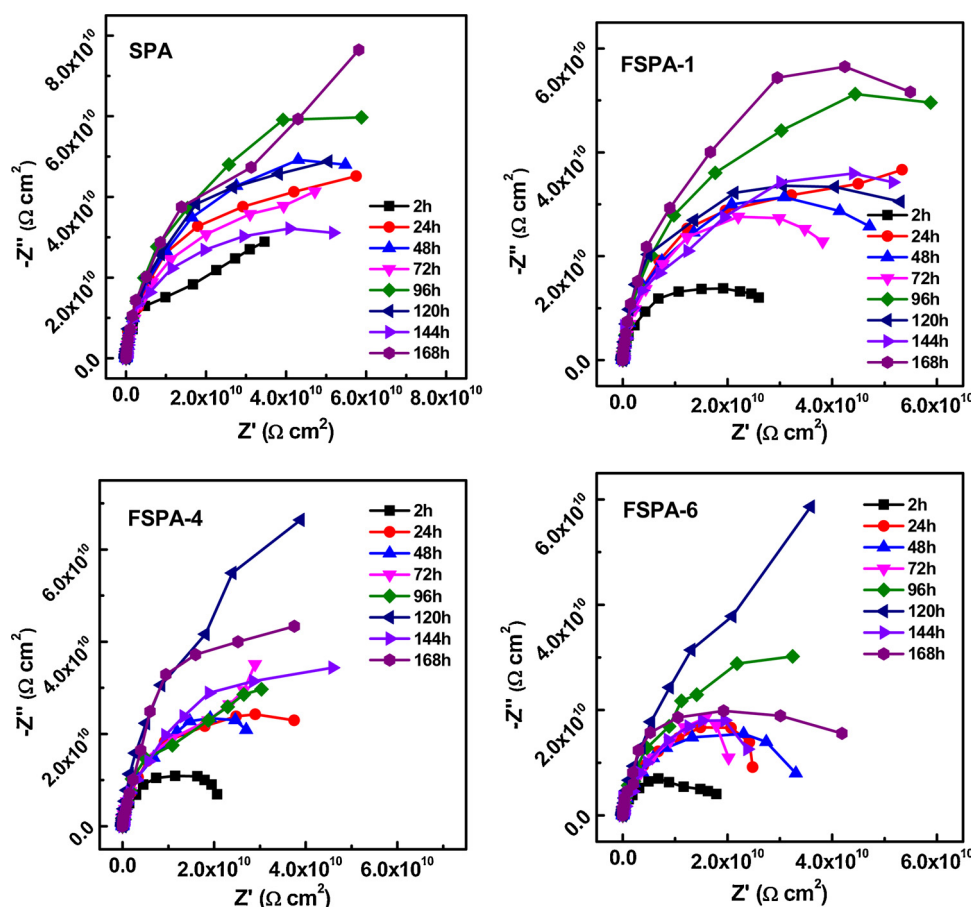


Fig. 5. Argand space plots of samples.

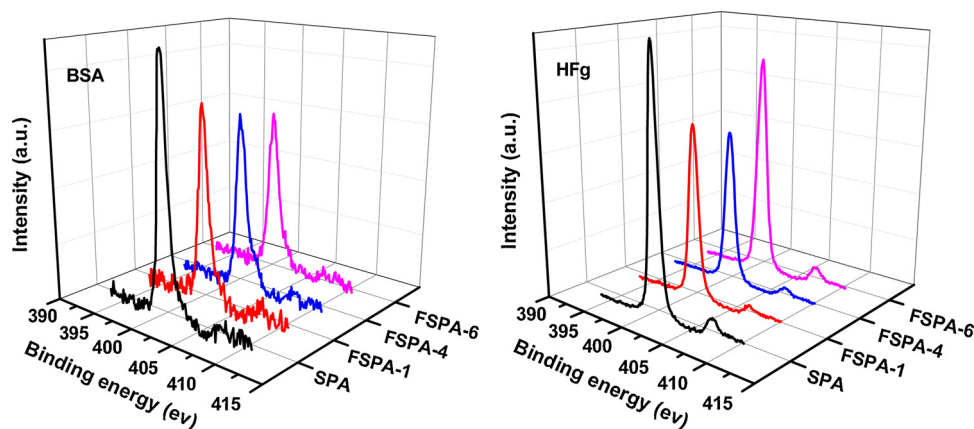
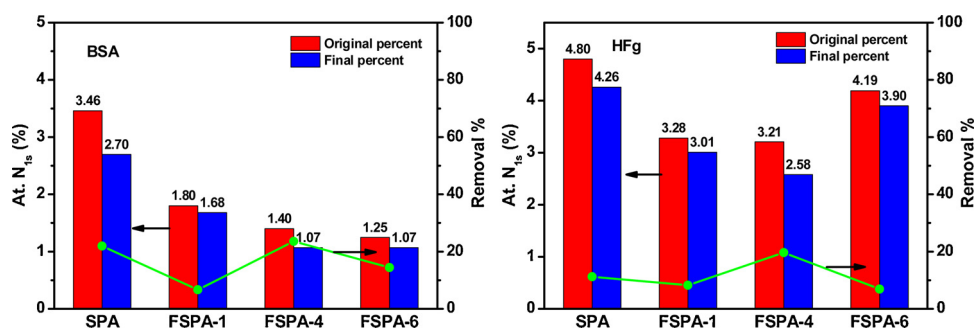
Fig. 7. XPS spectra of N_{1s} on polymer films surface after proteins adsorbed.

Fig. 8. Plot of proteins adsorbed and percent removal of proteins.

structure of the polymer to a certain degree, preventing the rearrangement of surface fluorinated groups.

Real impedance (Z') at low frequency region ($f = 0.01$ Hz) in the EIS measurement was used to evaluate the anticorrosion performance of coatings [25,45,46], higher $Z'|_{0.01 \text{ Hz}}$ means lower corrosion rate [47]. It is generally believed that the organic coating possesses good anticorrosion performance when $Z'|_{0.01 \text{ Hz}}$ is equal to, or greater than $10^6 \Omega\text{cm}^2$ and which can be used for industrial applications [45]. Whilst the coatings with a $Z'|_{0.01 \text{ Hz}}$ being larger than $10^8 \Omega\text{cm}^2$ have excellent anticorrosion performance [48]. Fig. 5 presents the Argand space plots of SPA, FSPA-1, FSPA-4 and FSPA-6 after being immersed in 3.5 wt% NaCl solution for 7 d. Obviously, all $Z'|_{0.01 \text{ Hz}}$ values of the samples after immersion for 7 d are larger than $10^{10} \Omega\text{cm}^2$, indicating their excellent anticorrosion performances, which is attributed to the dense surface formed by cross-linking between polymer molecules. According to the physical definition of resistance (eq. 1), the coating's thickness (d), except for the electrode working area (S), also has a great influence on $Z'|_{0.01 \text{ Hz}}$.

$$R = \frac{\rho \cdot d}{S} \quad (1)$$

where R and ρ are the coating's resistance and electrical resistivity, respectively. Consequently, it is difficult to compare the anticorrosion performance of coatings (with the same working area but with different thickness) by simply comparing their $Z'|_{0.01 \text{ Hz}}$ values. Therefore, here the $Z'|_{0.01 \text{ Hz}}$ value per unit area and per unit thickness (designated as R_{sd} , hereinafter) is taken into account. The evolution of R_{sd} with immersion time (Fig. 6) clearly shows that the R_{sd} values of FSPA-1 (with 4.3 wt% FMA) and SPA are much close, and both of them have a similar evolution trend with the immersion time. Although the fluorinated groups on the polymer surface will be rearranged in water [44], resulting in the more and more chaotic fluctuations of R_{sd} with the increase of FMA, R_{sd} values of the high fluorine content polymers (FSPA-4 and FSPA-6) are still significantly larger than that of SPA. Especially, all

R_{sd} values of the investigated polymers are larger than $10^8 \Omega\text{cm}^2 \mu\text{m}^{-1}$ even after immersion for 7 d, foreboding their excellent anticorrosion performance.

3.5. Protein adsorption and release on polymers surface

Quantitative measurement of protein adsorption on the surface is not an easy task [5]. Conventional methods for quantitative measurement of protein adsorption mainly include surface plasmon resonance (SPR) [49,50], quartz crystal microbalance (QCM) [51,52] and XPS [24,53,54]. SPR and QCM have their own assumptions which responses to the specific requirements of the amount of protein adsorbed on the prepared samples [5]. When there is no nitrogen atom in the sample, XPS is a simple and effective method for quantitatively evaluating the adsorption of proteins based on the measurement of N_{1s} peak at 400 eV [24,53,54]. Here, bovine serum albumin (BSA) and human plasma fibrinogen (HFg) were used as model proteins to evaluate proteins adsorption performance on the surfaces of SPA, FSPA-1, FSPA-4 and FSPA-6. The relative intensities of the N_{1s} XPS peaks (Fig. 7) clearly show that the fluorinated polymers (FSPAs) adsorb less amount of protein than the SPA. Fig. 8 shows the percentage of proteins adsorbed on the polymers surface before and after washing, and their corresponding percent removals from polymers surface.

It can be seen that the surface adsorption of the two proteins decreased with the increase of FMA content except for the adsorption of HFg on FSPA-6. BSA achieves its lowest adsorption when FMA content is around 26.7 wt%, while HFg achieves its lowest when FMA content is around 15.4 wt%. That is to say, minimum adsorption amount of the two proteins appears on surface of the polymers with different content of fluorinated groups. Furthermore, when the FMA content is approximately 15.4 wt%, the removal of BSA and HFg at a water channel pressure of ~ 85 Pa for 2 s are 23.6 % and 19.6 %, respectively, and each of them reaches their maximum. The above results indicate that FSPA-4 has the lowest adsorption and the maximum removal of the

proteins, inferring its better antifouling performance than other samples.

4. Conclusions

A series of fluorinated/silanized polyacrylate amphiphilic polymers have been successfully synthesized via a two-step free radical initiated polyreaction, and characterized especially for their anticorrosion and antifouling performances. Whilst, a modified corrosion resistance (R_{sa}), which takes the polymer film thickness into account, was proposed and adopted to compare the polymer's anticorrosion performance. The partial cross-linking between the polymer molecules gives the polymer films good stability, excellent adhesion strength, and high hardness and corrosion resistance. By altering the content of FMA, the surface wettability and protein adsorption performance of the polymer films can be significantly regulated. Polymer FSPA-4 with the content of FMA being 15.4 wt% possesses the optimal anti-protein adsorption property.

CRediT authorship contribution statement

Benfeng Zhu: Conceptualization, Methodology, Software, Data curation, Writing - original draft. **Zehan Liu:** Investigation, Methodology. **Jiao Liu:** Conceptualization, Data curation. **Yumeng Yang:** Methodology, Software. **Yanbin Meng:** Methodology, Software. **Fang Yu:** Methodology, Software. **Li Jiang:** Formal analysis. **Guoying Wei:** Formal analysis. **Zhao Zhang:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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